Speaker Information

[Name] Yusuke Nasu

[Institute] Institute of Biological Chemistry, Academia Sinica

[Position] Assistant Research Fellow

[Email Address] nasu@as.edu.tw

[Lab / Personal Website] https://nasulab.org/

Talk Information

Genetically encoded fluorescent biosensors for neuroscience

At the core of our laboratory's work is the pioneering field of protein engineering, aimed at developing functional proteins with exceptional properties. A key focus is the creation of fluorescent biosensors that can reversibly alter their fluorescence intensity in response to specific targets. To further our research goals, we collaborate with a global network of experts in neuroscience. Through these partnerships, we aim to accelerate scientific discovery and enhance our understanding of critical biological processes. In this seminar, we will introduce our technology and the state-of-the-art biosensors.

Research Overview (for collaboration discussion)

[Research Interests] Protein engineering for fluorescent biosensors

[Current Research Projects] Development of genetically encoded biosensors for neuronal metabolism

[Techniques] protein engineering

[Aspects or techniques of interest for future collaboration] application of biosensors in living model organisms to investigate metabolism in neurons and glial cells with spatiotemporal resolution

Short Bio Yusuke Nasu obtained his BSc in chemistry from the University of Tokyo in 2009 and his PhD in analytical chemistry from the University of Tokyo in 2016. He subsequently worked on analytical method development for pharmaceuticals at Chugai Pharmaceutical Co., Ltd. (Roche Group). He then held postdoctoral positions at the University of Alberta and the University of Tokyo, where he specialized in directed protein evolution for the development of fluorescent biosensors. In 2024, he established his independent research group at the Institute of Biological Chemistry, Academia Sinica, where his research focuses on deepening and broadening protein engineering approaches for fluorescent biosensors.